

LONG ISLAND UNIVERSITY · LEWYT COLLEGE OF VETERINARY MEDICINE

DVM 612 · PRINCIPLES OF SURGERY

Preoperative Evaluation, Patient Preparation & Postoperative Care

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Lecture | 38 slides | 60 minutes

Exam, labs, then you write the ASA. Then prep. Then 24 hours of monitoring and complication management.

Required: Fossum 2018; Hendrickson & Baird 2013.

Also used: ECFVG CPE Manual of Administration 2026, Anesthesia and Surgery.

Learning objectives

- ▶ Assign American Society of Anesthesiologists (ASA) status after today's PE and labs. Correct deficits. Write premed, induction, MAC, tube, bag, and fluid rate.
- ▶ If albumin is low: find the cause, delay elective if you can, and do not flood with crystalloid.
- ▶ Prep for aseptic surgery: clip, dirty scrub, OR attire, full scrub, drape, instrument table, closed glove.
- ▶ Name a break in asepsis and correct it, before or after the incision. Classify the wound: clean, clean-contaminated, contaminated, or dirty.
- ▶ Write the postop plan, the surgical report, and 24-hour emergency criteria.

Exam → labs → ASA → ready for prep → clip, dirty scrub, OR, full scrub, drape, instruments → first 24 hours.

Prepare the patient and yourself

Causes of SSI

Hair, skin flora, hypothermia, poor hemostasis, dead space, and breaks in asepsis.

Client-owned patients

Elective OHE patients go home to the owner after recovery. Documentation and the discharge conversation are part of surgical care.

Set before you cut

Analgesia, antibiotics, temperature management, and client expectations are set in the preoperative period.

Surgeon responsibilities

- ▶ Prepare the patient for a surgical procedure.
- ▶ Prepare yourself for a surgical procedure.
- ▶ Perform the procedure. Then monitor the first 24 hours: a weak ligature or a weak linea is a technical failure you must recognize (CPE MOA 2026 Surgery).

Halsted's principles

1. Gentle tissue handling

Prep trauma, clipper burn, and crushing towel clamps are tissue handling.

2. Meticulous hemostasis

Preop coagulopathy; postop hemorrhage is a 24-hour emergency.

3. Preserve blood supply

Place clamps and bandages so skin perfusion is preserved.

4. Strict asepsis

Patient preparation, surgeon preparation, and draping.

5. No tension

Plan incision location and closure now, not after the fascia splits.

6. Accurate apposition

Postop incision care protects the apposition you create.

7. Obliterate dead space

Plan dead space now: sutures, bandage, or a drain with a written pull plan.

The perioperative continuum

PRE-OP

PE, labs, ASA
Correct deficits
Premed, induce, MAC
ETT, bag, fluids



PREP

Clip, dirty scrub
Shoe covers, cap, mask
OR: full scrub, gown
Drape, instruments



INTRA-OP

Mark incision
Hold asepsis
Halsted, temp, pain
Chart every 5–10 min



POST-OP

Airway, pain, heat
Watch 24 hours
Report + discharge
Recheck plan

Anesthesia clock · ready for surgical prep

From the preoperative exam: airway in and secured, machine on, patent IV running, surgical plane, monitoring started. Then call for prep (CPE MOA 2026 Anesthesia).

Surgery clock · last skin suture

From the start of patient preparation to the last skin suture, including re-gowning. Then 24 hours of monitoring and complication management (CPE MOA 2026 Surgery).

Preoperative physical examination

T • HR • RR

Write the numbers. Conscious dog: HR 70–140, RR 20–40. Cat: HR 145–200, RR 20–40 (Fossum).

mm / CRT

Mucous membrane color and capillary refill. Pink vs pale vs injected. CRT in seconds.

Heart and lungs

Auscultate both sides. Record any murmur, arrhythmia, or crackles before you choose drugs.

Hydration + site

Skin tent, tacky mm. Look at the surgical site: pyoderma, heat, pain. Abdomen if you might open it.

Minimum labs on the anesthesia record • PCV, TP, BUN (CPE MOA 2026 Appendix 3)

PCV

Oxygen-carrying capacity.
Anemia vs hemoconcentration.
With TP: dehydration vs blood loss.

TP

Protein / oncotic pressure. High with PCV: dehydration. Low: loss, liver, or dilution. Changes fluids and healing.

BUN

Kidney, hydration, GI bleed.
Azotemia: skip NSAIDs, rethink fluids, maybe do not cut today.

American Society of Anesthesiologists (ASA)

ASA status	Definition	Example
Status 1	Normal healthy patient	Healthy patient for neutering
Status 2	Patient with mild systemic disease	Well compensated mild mitral degeneration
Status 3	Patient with moderate systemic disease	Cat with CKD, IRIS stage 3
Status 4	Severe systemic disease; constant threat to life	Hemoabdomen, bleeding splenic mass
Status 5	Moribund; not expected to survive without the operation	Hypotensive, hypothermic, obtunded septic cat

E = emergency. Status 3 is moderate and compensated. Status 4 is uncompensated and a threat to life. Today's PE and labs write the number. A normal CBC does not write Status 1.

MoMo: today's PE



SF DSH, 6 yr, 4.25 kg, BCS 5/9. Acute vomiting and lethargy this morning. Two vomits (food, then liquid). Construction at home. T 98.0 °F, HR 200, RR 30, mm pink tacky, CRT <2 s, QAR. Do not write the ASA yet.

Why this looked surgical

Possible FB or toxin was on the list. Mildly enlarged abdomen. That is how cats get booked for an exploratory.

Abdomen and the rest of PE

Mildly enlarged abdomen on palpation. Rectal not examined. The PE did not prove a foreign body.

Also on PE

Eyes, ears, nose normal. Oral pink, CRT <2. Lungs clear, eupneic. No murmur, regular rhythm. Ambulatory ×4. LN and skin WNL. Neuro: normal gait, CNs intact. MCS 3/3. Generalized small stature.

MoMo: labs. Write the ASA

HCT

44% (28–50)

TP (total protein)

5.1 g/dL

BUN / creatinine

BUN 48.7 → 100 (15–32). Cr 3.0 → 4.71.

Out of range

WBC 26 (3–16).
Neutrophils 24 (4–11). Na 142. BE –7.4.
Glucose 136. Phos high; no number printed.

Quiz. Assign ASA. Add E if emergency. Write it before you hear the key. AUS: right kidney fluid-filled, non-functional; left kidney reduced corticomedullary architecture. HCT in range does not write Status 1.

1. Status?

Number, and E or not.

2. Why not Status 3?

Is this still compensated CKD, or a constant threat to life?

3. Cut today?

Exploratory on vomiting and an enlarged abdomen?

Willie: today's PE

MN Cavalier, 6 y 11 mo, 13.7 kg. Acute ataxia and disorientation ~1 hour. Found on the floor; fell off the couch twice. Cytopoint. T 100.8 °F, HR 132, RR 52, mm pink, CRT 2 s, dull/quiet. Do not write the ASA yet.

Left ear (AS)

Brown and bloody discharge.
Pedal reflex at the pinna base.
Canal patent. Cartilage
hardened. Tympanic membrane
visible but swollen.

Vestibular

Head tilt / leaning toward the
left (infected ear, AS). Stumbling.
Circling or tight turns to the left.
Nystagmus: rapid left, slow
right.

Wheelbarrow / CP

Wheelbarrow: left intact, right
absent. Prolonged
proprioception, right limbs.
Knuckling on the right limbs
when walking. Limbs slip.

Also on PE: Grade II/VI left systolic murmur. Lungs clear, eupneic. Heavy tartar. Yellow crusts on the nasal philtrum. Eyes: PLR present. Next: today's chemistry and CBC. Then you write the ASA. Owner identifiers stay off these slides. Do not use a drug list on this dog.

Willie: labs. Write the ASA

HCT

47.5% (36.9–60.0)

TP

6.4 g/dL (5.5–7.6)

BUN / creatinine

BUN 11.9 (9.0–29.0). Cr
0.7 (0.4–1.4).

Out of range

Phos 6.0 (1.9–5.0).
Glucose 130 (75–
125). CBC in range.

Quiz. Assign ASA. Add E if emergency. Write it before you hear the key. HCT in range does not write Status 1.

1. Status?

Number, and E or not.

2. Why not Status 1?

What on today's PE is not a normal healthy patient?

3. Why not Status 4?

Is this a constant threat to life today?

Pre-anesthetic assessment

PCV / TP: EDTA tube → spin hematocrit → read PCV → snap the capillary tube → refractometer TP. BUN: chemistry or Azo stick. Request, record, interpret. Then ASA. Then premed IM/SQ. Not before the exam.

Field	MoMo (quiz)	Healthy Lab OHE	Willie (quiz)
Weight	4.25 kg	25 kg	13.7 kg
T · HR · RR	98.0 °F · 200 · 30	normal TPR	100.8 °F · 132 · 52
mm / CRT	pink, tacky / <2 s	normal	pink / 2 s
Other PE	Mild abdominal enlargement Ambulatory ×4. No FB proven	Elective OHE normal PE	AS otitis. Head tilt L Circling L. R knuckling
PCV / HCT	HCT 44% (28–50)	within reference	HCT 47.5% (36.9–60.0)
TP	5.1 g/dL (total protein)	within reference	6.4 g/dL (5.5–7.6)
BUN	48.7 → 100 Cr 3.0 → 4.71	within reference	11.9 · Cr 0.7 Phos 6.0 · Glu 130
ASA	write it	1	write it

Laboratory values inform risk

Finding	Typical healthy adult	Repeat / stabilize first
PCV	Within reference	Anemia: use severity and chronicity (Grubb 2020)
TP	Within reference	High with PCV: dehydration. Low: loss / dilution
BUN	Within reference	Azotemia: skip NSAID, rethink fluids and the cut
Potassium	Within reference	>6.0 mEq/L (Grubb 2020: correct first)
Albumin	Within reference	<2.0 g/dL: edema, poor healing (Pardo 2024)

MOA minimum is PCV, TP, BUN. A normal CBC does not assign Status 1. Azotemia can cancel the cut. Low albumin is a deficit, not an ASA number.

Anesthetic drug protocol

1. Premed after ASA

IM or SQ. Not before the exam.
Hospital: DKT or BAA. Same columns, hospital doses.

2. Induce IV to effect

Alfaxalone 10 mg/mL. Draw 5.5 mL.
Give until you can intubate.

3. Iso or sevo to MAC

Premed, if you add it, lowers MAC and lowers alfaxalone.

Anesthetic drug protocol

25 kg · ASA 1 · elective OHE

	Drug	Concentration	Dose	Volume	Route
Analgesic	Carprofen (Rimadyl)	50 mg / mL	4.4 mg / kg	2.2 mL (110 mg)	SQ ~2 h before
Induction	Alfaxalone (Alfaxan)	10 mg / mL	IV to effect	draw 5.5 mL	IV
Inhalant	Isoflurane	to MAC	to effect	vaporizer	inhalant

IV fluids	Name: LRS Rate: 125 mL/hr (25 × 5)	Why these drugs Carprofen: 25 × 4.4 = 110 mg. 110 / 50 = 2.2 mL SQ (Rimadyl). Alfaxalone: draw 25 × 2.2 = 55 mg = 5.5 mL. Give IV to effect (Alfaxan). 2.2 mg/kg is the unpremedicated field-study mean, not a fixed bolus. Cat OHE: Onsior 2 mg/kg SQ. Do not stack an NSAID with a steroid.
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Sedation quality 1-5	Induction quality 1-5	Circle inhalant
1 2 3 4 5	1 2 3 4 5	Isoflurane Sevoflurane
Circle circuit	ETT · bag (if intubated)	Fresh-gas flow
Rebreathing Non-rebreathing	10 mm 2 L circle	2-3 L/min, then ≥ 0.5 L/min

Correct deficits before you cut

Correct these first

Grubb et al. 2020, before anesthesia:

Dehydration / hypovolemia

K⁺ >6.0 mEq/L

pH <7.2

Glucose <60 mg/dL

Hypothermia <99 °F

Anemia (severity, chronicity)

Life-threatening arrhythmia

CHF, cyanosis, pneumothorax

Oliguria / anuria

Uncontrolled pain or anxiety

If albumin is low

Find the cause: GI loss, kidney, liver, exudate, starvation.

Delay elective surgery if you can treat and feed.

<2.0 g/dL: edema, poor perfusion, poor wound healing (Pardo 2024).

Plasma: about 20–25 mL/kg to raise albumin 0.5 g/dL. Often impractical.

Species-specific albumin if severe and you must operate. Human albumin can cause anaphylaxis.

Do not flood with crystalloid when the problem is oncotic pressure.

If you still operate

Delay elective for pyoderma over the site.

Emergency: resuscitate volume, then cut. You cannot wait for albumin to normalize in GDV or septic abdomen.

Gentle tissue handling. Obliterate dead space. Warm the patient. Nutrition as soon as the gut allows.

Rising creatinine that threatens life: cancel the elective or exploratory cut.

Fasting and informed consent

Fasting (Grubb et al. 2020 AAHA)

- ▶ Healthy adult dog/cat: food 4–6 h; water until premedication.
- ▶ Some hospitals still use 8–12 h NPO. Teach aspiration versus hypoglycemia.
- ▶ Neonates / <2 kg: food fast no longer than 1–2 h.
- ▶ Brachycephalics: same fasting as the adult or neonate row. Pre-oxygenate 3 min with 100% O₂ (Grubb 2020). Extubate when they can protect the airway, not at the first swallow.
- ▶ Ask what was eaten this morning. Clients feed 'just a biscuit.'
- ▶ Diabetics: write the insulin dose and a small meal with anesthesia before drop-off.

Consent, then file it

- ▶ Procedure name in plain language, and the reason.
- ▶ Benefits, alternatives including medical management, and what happens if we wait.
- ▶ Material risks in one sentence each: anesthesia death, hemorrhage, infection, dehiscence. After OHE, name urinary incontinence.
- ▶ Estimate: what is included and what is not.
- ▶ Resuscitation code / DNR before induction.
- ▶ File the signed form. Then start induction.

ASA, tube, bag, and fluid rate

Endotracheal tube

$ID\text{ mm} = (kg / 4) + 3.5$

Dog start. ±0.5 mm ready. Largest through arytenoids (Grubb 2020). Cat: 3.0–4.5 mm (not the dog formula).

Reservoir bag

$mL = kg \times 60, \text{ round UP}$

Round up to 0.5, 1, 2, or 3 L. NRC if <3–5 kg. Circle if larger. Leak-test, then OPEN the pop-off.

IV fluid rate

$\text{Dog } kg \times 5 \cdot \text{cat } kg \times 3\text{--}5$

mL/hr. Balanced crystalloid (Pardo 2024). Not 10 mL/kg/hr. Example: $20 \times 5 = 100\text{ mL/hr}$.

Patient	kg	ASA	ETT	Bag / circuit	IV fluids	Fresh-gas flow
Example 20 kg dog	20	1	8.5 mm 8.0 / 9.0 ready	2 L circle 1.2 L → 2 L	LRS 100 mL/hr 20 × 5	2–3 L/min, then ≥ 0.5 L/min
Healthy Lab elective OHE	25	1	10 mm 9.5 / 10.5 ready	2 L circle 1.5 L → 2 L	LRS 125 mL/hr 25 × 5	2–3 L/min, then 0.5–1 L/min
MoMo math only	4.25	4-E	3.5–4.0 mm cat; not dog formula	0.5 L NRC 255 mL → 0.5 L	LRS 13 mL/hr 4.25 × 3	0.85–1.7 L/min 200–400 mL/kg/min

MoMo's perioperative record

Perioperative record

DVM 612 · Preoperative evaluation and recovery · teaching form, not a hospital original

Complete this page before the first drug

Patient	Species / breed	Sex / age	Weight	ASA	What you planned today
MoMo	Feline · DSH	SF · 6 yr	4.25 kg BCS 5/9	4-E	Exploratory considered. Then cancelled.

PREOPERATIVE EVALUATION · complete before any drug

T °F	HR	RR	mm / CRT	Mentation	Heart / lungs
98.0	200	30	pink, tacky / <2 s	QAR	NSR; no murmur; eupneic
PCV		TP		BUN	Other labs
44%		5.1		48.7 → 100	Cr 3.0 → 4.71. WBC 26. NEU 24. HCT 44%. TP 5.1. BE -7.4. Na 142.

What changes the plan	Two vomits. Mildly enlarged abdomen. PE did not prove FB. AUS: right kidney fluid-filled, non-functional; left kidney reduced CM architecture. HCT 44%. TP 5.1. BUN 48.7 → 100. Cr 3.0 → 4.71. Do not clip.
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PRE-OP BOXES · before the first drug

☒ ASA Status 4-E after labs and imaging

☒ POCUS / AUS before any clippers

☒ Right kidney fluid-filled, non-functional. Left kidney reduced CM architecture. HCT 44%. TP 5.1. BUN 48.7 → 100. Cr 3.0 → 4.71

☒ No clippers. No incision. No exploratory.

☒ NSAIDs contraindicated (azotemic cat)

☒ Consent includes medical care and euthanasia

☒ Write SURGERY CANCELLED on this record

RECOVERY / NEXT 24 HOURS · monitor and manage complications

☒ This is not a recovery from surgery

☒ Supportive care. Recheck kidneys.

☒ Right kidney fluid-filled and non-functional

☒ Progressive azotemia: creatinine 3.0 → 4.71 on fluids

☒ Offer medical care, referral, or euthanasia

☒ Write the decision on this record

☒ Owner elected humane euthanasia

SURGERY CANCELLED · RECORD THE DECISION

Do not cut until you know why the creatinine rose. Owner elected euthanasia.

Surgical wound classification

1. Clean

No infection. Hollow viscus not entered.

Elective OHE, orchiectomy, simple skin mass.

Skip routine prophylaxis (Frey 2022).

2. Clean-contaminated

GI, respiratory, or urinary tract entered under control.
Minor break in asepsis.

Cystotomy. Enterotomy without spillage.

3. Contaminated

Fresh open trauma. Major viscus spillage. Major break in sterile technique.

Time IV prophylaxis. Fix the break.

4. Dirty / infected

Pus, abscess, old trauma, perforated viscus with peritonitis.

Infected ear canal. Pyometra with pus.

This is treatment, not prophylaxis.

Write the class first. Clean elective: skip (Frey 2022 AAFP/AAHA stewardship). When prophylaxis is indicated: 30–60 min before incision; cefazolin 22 mg/kg IV (Gonzalez 2017, extra-label). Redose if surgery lasts >90 min (Whittem 1999). Stop at closure unless you are treating infection.

Clip, dirty scrub, then the OR

Patient and surgeon preparation

Clip and dirty-scrub in the prep area. Shoe covers, cap, and mask before the door. Full scrub, drape, and instruments in the OR.

1	Ready for prep ETT in. Cuff holds. IV running. Surgical plane. Then you may clip.	2	Clip #40 blade. Vacuum the hair. Express the bladder if you will open the abdomen.	3	Dirty scrub Preliminary antiseptic in the prep area. This is not the sterile prep.	4	Attire at the door Shoe covers, hair cap, mask. Then enter the OR. Gown is not hallway wear.	5	Move and position Into the OR. Pad. Ties. Pulse distal to each tie. Final check.
6	Full sterile scrub In the room, after positioning. Clock contact time. Spiral out. Drop the sponge.	7	Gown, closed glove Surgical hand scrub first. Hands stay in the cuffs. Then glove.	8	Drape Four-quadrant towels, then the large drape. Only the incision is in the window.	9	Instrument setup Open the pack. Check the indicator. Arrange. Count sponges and instruments.	10	Timeout. Mark incision Announce the incision. Mark first cut on the chart. Clock runs to the last skin suture.

Hair removal

Too narrow. Hair at the margin.



24 hours after OHE. Clip was wide enough.



Celiotomy: ~4 cm cranial to the xiphoid, ~4 cm caudal to the pubis, 3–4 cm past the nipples (Thieman Mankin 2019). A 4 kg cat does not have 20 cm of extra body wall. Wide enough to extend or place a drain. Warm the patient (Grubb: hypothermia <98 °F).

Photos: Uwe Gille, CC0; Liannadavis, CC BY-SA 4.0.

Skin antiseptics

Human Betadine Solution = 10% PVP-I. Veterinary paint = 5%, not 10%.

Eye: 0.2% only. 2 min scrub + 2 min soak. Do not put 10% or 5% undiluted on the cornea (Roberts 1986).

CHG · Nolvasan

2% chlorhexidine acetate
MIN contact: 2 to 4 min
Keep out of eyes
Trunk / intact skin
Alcohol: rinse, then dry

Skin · Betadine vet

7.5% scrub
MIN contact: about 5 min
Rinse with sterile water
Paint 5% solution
Dry, then drape

Eye · Roberts 1986

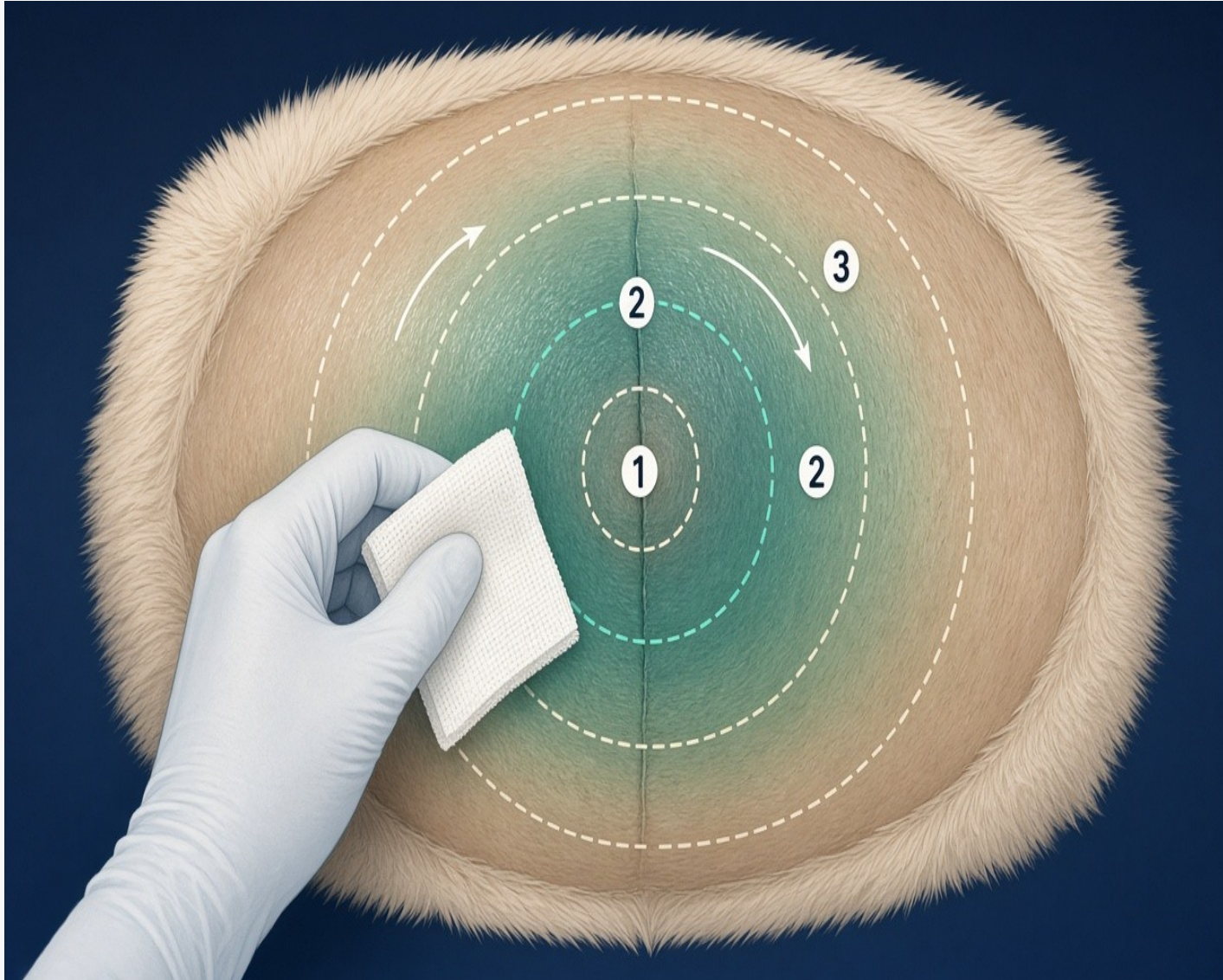
0.2%: 2 min scrub + 2 min soak
10% stock → dilute 1:50
5% bottle → dilute 1:25
Not 10% or 5% undiluted
Perforation: saline only

Nolvasan Surgical Scrub: 2% chlorhexidine acetate, DailyMed NDC 54771-8701, wash 2 to 4 min, keep out of eyes.

BETADINE Surgical Scrub Veterinary NDC 67618-154 (7.5%, lather about 5 min). BETADINE Solution Veterinary NDC 67618-155 (5% paint, not 10%).

Eye: Roberts 1986, healthy cornea. 1:50 of 10% = 0.2%, 2-min scrub + 2-min soak. Do not use undiluted 10% or undiluted 5% paint. 1:2 caused edema in 1/15 eyes. Perforation: saline.

Patient skin preparation technique



Technique

- ▶ Dirty (preliminary) scrub in the prep area, after the clip.
- ▶ Move to the OR. Position, pad, tie.
- ▶ Full (sterile) scrub in the room. Start at the incision. Spiral out. Drop the sponge.
- ▶ Clock the minimum contact time.
- ▶ Skin: 7.5% ~5 min, then 5% paint.
- ▶ Eye: 0.2% (10% 1:50 or 5% 1:25). 2 min + 2 min. Not 10% or 5% undiluted. Perforation: saline only.

Patient positioning



Visible in this photograph

- ET tube + pulse ox
- IV catheter + fluids
- Circle + reservoir bag
- V-trough / padding
- Ties snug; pulse distal to each tie
- Clip after a surgical plane

Tube tip: midway larynx to thoracic inlet (Grubb 2020).
Confirm: ETCO₂, bag move, no esophageal tube.

Photo: Anja, CC BY-SA 4.0.

Draping

OHE: hair visible at the drape edge



The window: clipped skin only



Near towel first. Four towels box the field. Towel clamps on skin, not through the drape to viscera. Large drape last. Only the incision is in the window. Hair stays under the drape. Re-clip until the window is hair-free. Photos: Cpl. Samuel A. Nasso, USMC, public domain; MC1 Ryan D. McLearnon, USN, public domain.

OR attire, gown, glove, instruments



Closed gloving: both hands remain inside the gown cuffs; the right cuff is used to pull the glove over the left hand.

Before you cut

- ▶ Door: shoe covers, hair cap, mask. Then enter.
- ▶ Surgical hand scrub. Gown. Closed glove (hands stay in the cuffs).
- ▶ Drape. Then open the instrument pack. Check the indicator.
- ▶ Arrange. Count sponges and instruments. Mayo stand.
- ▶ Name the monitors: ECG, SpO2, ETCO2, temp, fluids.

Left: closed-gloving technique (hands stay inside the gown cuffs). Right: door attire, then gown, drape, open the pack, count, Mayo stand. Timeout. Announce the incision and mark it on the chart.

Breaks in asepsis

If you contaminate

Prep, gown, glove, drape: say it immediately. Re-glove, re-gown, or re-drape. Do not continue with a contaminated field.

After the incision: notice, announce, and fully correct. Sleeve in the abdomen, instrument off the table, hole in a glove: stop and fix it.

Protect the patient

1. Control significant hemorrhage.
2. Create a secure abdominal wall closure.
3. Achieve and maintain an aseptic field.
4. Identify and protect adjacent organs before you clamp.
5. Complete a sponge and instrument count before closure.

Surgical competence includes the first 24 hours of physiologic monitoring and complication management (CPE MOA 2026 Surgery). A weak ligature or a weak linea is a technical failure you must recognize. That is not infinite legal liability for every recovery-ward or owner event.

Knowledge check

A

When do you clip the OHE field?

After ready for prep. Clip and dirty-scrub in the prep area. Full sterile scrub in the OR after positioning.

B

How do you prep conjunctiva with iodine?

Intact cornea: 0.2% (10% 1:50 or 5% 1:25), 2 min scrub + 2 min soak. Not 10% or 5% undiluted.
Perforation: saline only.

C

Albumin 1.6 g/dL, elective mass. First move?

Delay. Find the cause. Feed. Do not crystalloid-flood. Do not cut today.

D

25 kg dog. Tube, bag, and IV fluid rate if you intubate?

10 mm, 2 L circle, LRS 125 mL/hr (5 mL/kg/hr).

Immediate recovery



Recovery priorities

- ▶ Extubate when swallow returns.
- ▶ Most anesthetic deaths are in recovery, often in the first 3 hours (Grubb 2020).
- ▶ Pale + tachycardic after celiotomy: return to OR.
- ▶ NSAID: carprofen in the dog, Onsior in the cat. Do not stack with a steroid.
- ▶ E-collar on before they can lick.

Anesthesia chart: start, incision, criteria

Anesthesia chart
Mark anesthesia start and first incision. Record every 5–10 min against the range.

What you plot	Normal operating range	Anesth. start	5 min	Incision	15 min	Last suture
Heart rate	dog 60–140 · cat 100–180					
Respiratory rate	dog 8–20 · cat 10–25					
BP SAP / MAP / DAP	SAP ≥90 · MAP ≥70 · DAP ≥40					
Isoflurane or sevoflurane %	to effect					
Oxygen flow L / min	2–3 L, then 20–40 mL / kg / min · min 0.5 L					
End-tidal CO2	40–50 mm Hg · PPV if >60					
SpO2	≥95%					
Temperature	≥98 °F (36.7 °C)					
Fluid rate mL / hr	dog 5 mL / kg / hr · cat 3–5					
Total fluid mL	rate × hours					
Times you mark		Recording criteria			If out of range, then sign	
Anesthesia start. First incision. Last skin suture. Extubation. Write the clock time in the gold columns.		Every 5–10 min, in real time. Plot HR, RR, BP, inhalant, O2, ETCO2, SpO2, temp, fluids against the range. Write the number you see.			Event, time, and correction. Fluids: type, rate, total mL. Recovery analgesic. Sign the record.	

Postop analgesia and incision care

On-label NSAID

- ▶ Score pain. Dogs: Glasgow CMPS-SF. Cats: Feline Grimace Scale. Opioid as planned.
- ▶ Dog: carprofen (Rimadyl) 4.4 mg/kg SQ. Dogs only. Give ~2 hours before incision so levels are on board (pre-emptive; Rimadyl label).
- ▶ Cat: Onsior (robenacoxib) 2 mg/kg SQ. Cats only. First dose ~30 min before incision so levels are on board. OHE, castration, orthopedic. Up to 3 days.
- ▶ Dog: carprofen. Cat: Onsior. Do not swap species in this course.
- ▶ Do not stack an NSAID with a steroid. Skip NSAID if the patient is azotemic.

Antibiotic injection and incision

- ▶ A postoperative antibiotic injection is treatment. Write the drug, dose, and why.
- ▶ Clean elective OHE: skip (Frey 2022). Do not send home 14 days of cephalexin.
- ▶ Dirty or infected: continue as therapy, not prophylaxis.
- ▶ Look twice daily: swelling, discharge, gapping, smell, heat.
- ▶ E-collar that stays on. Leash walks 14 days.
- ▶ Skin sutures typically 10–14 days.

Complications in the first 24 hours

Hemorrhage

Pale mm, tachycardia, distending abdomen, drip from incision, collapsing. Stabilize and return to OR.

Airway / aspiration

Stertor, crackles, regurg on the pillow. Especially brachycephalics and after opioids.

Hernia / evisceration

Linea failure. Emergency. Protect viscera with sterile moist towels, opioids, OR now.

Dehiscence (later)

Often day 3–5. Skin versus fascia: fascial dehiscence is the emergency.

SSI

Redness, pain, discharge, fever, usually after day 3. Culture. Open, lavage, and treat the organism you grow.

Seroma / self-trauma

Dead space + motion. Restrict activity. Bandage if needed. Leave the fluid unless it is infected.

Surgical report and discharge

Minimum report elements

- ▶ Date/time, patient ID, surgeon, anesthesia, ASA.
- ▶ Procedure name and side. Position, clip/prep, approach.
- ▶ Findings in order. Ligatures, implants, samples.
- ▶ Suture: layer, material, size, pattern.
- ▶ Hemostasis, blood loss, sponge count, complications.
- ▶ Postop plan: fluids, pain, feeding, recheck.

Discharge, verbal and written

- ▶ What we did, in one sentence.
- ▶ When to start food and water, and how much.
- ▶ Every medication: name, dose, time, with food?
- ▶ Incision care and 14-day leash walks.
- ▶ E-collar on except when directly watching.
- ▶ Next appointment. ER criteria. After-hours number.

Emergency criteria for discharge

Come now

Collapse. Pale gums.
Distended abdomen.
Unstoppable bleeding. Can't breathe. Can't urinate.
Evisceration. Uncontrolled pain.

Call today

Not eating by morning. A few vomits. Mild incision redness. Diarrhea. E-collar problems.

Expected

Sleepy tonight. Small bruise. Slightly tacky incision. Reduced appetite this evening.

Answer: MoMo is Status 4-E

MoMo · SF DSH · 4.25 kg · today's PE and today's labs write the status. Do not cut.

Disease	Uncompensated	Not Status 3	Do not cut
Acute vomiting and lethargy. Azotemia. Right kidney fluid-filled and non-functional. Left kidney reduced corticomedullary architecture.	BUN 48.7 → 100. Cr 3.0 → 4.71. BE −7.4. Na 142. HR 200, tacky mm. That is a constant threat to life.	Status 3 is moderate and compensated. A HCT of 44% in range does not write Status 1. This is not compensated CKD.	Exploratory cancelled. NSAIDs contraindicated. Owner elected euthanasia. Record SURGERY CANCELLED.

Answer: Willie is Status 3-E

Willie · Cavalier · 13.7 kg · today's PE and today's labs write the status

Disease	Compensated	Not Status 1	Not Status 4
AS otitis. Head tilt left. Circling left. Nystagmus rapid left, slow right. Wheelbarrow right absent. Knuckling, right limbs. Grade II/VI murmur.	Pink, CRT 2 s. BUN 11.9 and creatinine 0.7 in range. Not a constant threat to life today.	HCT 47.5% in range does not write Status 1. This neuro PE is not a normal healthy patient.	Phos 6.0 and glucose 130 are high. They do not write Status 4. Status 4 is a constant threat to life.

Key points

1

Pre-anesthetic assessment: TPR, mm/CRT, then PCV, TP, BUN. Interpret. Circle ASA. Then IM or SQ premed.

2

Write drug, concentration, dose, volume, and route. Then IV fluid name and rate, ETT size, circuit, and fresh-gas flow.

3

Ready for surgical prep: airway secured, machine on, patent IV running, surgical plane, monitoring started.

4

Patient: clip, dirty scrub, move, full scrub, drape. Surgeon: shoe covers, cap, mask, then gown and closed glove. Instrument table, then announce incision.

5

Mark anesthesia start and first incision. Fill the chart every 5–10 min against the operating range. Then complications, recovery analgesics, and sign.

6

First 24 hours: monitor and manage complications. A weak ligature or a weak linea is still your technical problem.

Questions

When is the patient ready for clippers?

What do you write before the first IM or SQ premed?

References (course texts, guidelines, labels, and papers used in this hour)

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